

PAPER_228: Westerlund 2 Super Star Cluster " MUGE with High-Density OB-Supergiant Wind Feedback (10²¹; LMC)

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Session: 58 (grokshare8d951e12.txt extraction " Doc 6)

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Author: Daniel T. Murphy Framework: UQFF v4.8 (Star-Magic) Session: 58 (grokshare8d951e12.txt extraction " Doc 6) Date: March 2026 Classification: Novel MUGE " Extreme OB-Supergiant Wind Density Comparison Status: Proof-Quality Whitepaper

Abstract

Westerlund 2, the most massive super star cluster in the Milky Way (Carina-Sagittarius Arm, ~10 kly), is modelled with a 9-term MUGE. The distinguishing novelty is the OB-supergiant stellar wind density $\rho_{\text{wind}} = 10^{-20}$ kg/m³ " ten times denser than the LMC Tapestry (PAPER_227). This quantitative distinction within the shared $a_{\text{wind}} = \rho_{\text{wind}} v_{\text{wind}}^2 / \rho_{\text{fluid}}$ framework makes Westerlund 2 the highest-wind-density entry in the MUGE stellar-wind family.

1. Physical System

Westerlund 2 (Wd2) contains ~300 massive O/B stars in a ~ 4 ly core:

Parameter	Value
Distance	~ 10 kly (Milky Way, Carina arm)
M_{init}	$30,000 M_{\odot}$
M_{gas}	$100,000 M_{\odot}$
$M_{\text{dot_factor}}$	$M_{\text{gas}}/M_{\text{init}} = 3.33$
r	10 ly
B	10^{-10} T
τ_{SF}	2 Myr (rapid OB-dominated SF)
ρ_{wind}	10^{-20} kg/m^3
v_{wind}	2000 km/s

2. Key Distinction: Wind Density Factor

Comparative Table

System	M_{init}	ρ_{wind}	τ_{SF}	a_{wind} (m/s ²)
Tapestry LMC (PAPER_227)	$240 M_{\odot}$	10^{-21} kg/m^3	5 Myr	4×10^3

System	M_{init}	ρ_{wind}	τ_{SF}	a_{wind} (m/s ²)
Westerlund 2 (PAPER_228)	30{,}000 $M_{\odot}$	10^{-20} kg/m ³	2 Myr	4×10^4
Ratio	125 $\times$	10 $\times$	0.4 $\times$	10 $\times$

The 10 $\text{\AA}$ — higher wind density in Westerlund 2 directly scales the wind acceleration, making it the most wind-dominated system in this MUGE subset.

3. Gas-Accreting Mass (shared with PAPER_227)

$$M(t) = M_{\text{init}} \left(1 + \frac{M_{\text{gas}}}{M_{\text{init}}} e^{-t/\tau_{\text{SF}}} \right)$$

$$\dot{M}_{\text{UQFF}} = \dot{M}_0 \left(1 - [SSq] e^{-\kappa \Delta t} \right), \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$$

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$$\text{Name}_{\text{UQFF}}(r) = g_{\text{MUGE}}(r) \cdot \text{Bigl}(1 - [SSq] \cdot U_{\text{bi}} \wedge F_U(r,t) \text{Bigr}), \quad [SSq] = 0.57$$

With $M_{\text{dot_factor}} = 3.33$ (vs 41.7 for Tapestry): smaller fractional gas reservoir relative to the massive cluster.

4. Physical Rationale for Dense Wind

Westerlund 2 hosts ~ 300 O- and B-type supergiants, including WR 20a ($a \sim 83 M_{\odot} + 82 M_{\odot}$ WN binary). These stars produce winds with mass-loss rates $\dot{M}_{\text{star}} \approx 10^{-5}$ to $10^{-4} M_{\odot}/\text{yr}$. The collective wind deposits $\sim 10^{-20}$ kg/m³ in the 10 ly shell around the cluster core.

5. Calculator Class

```
class Westerlund2MUGESTellarWindCalculator(_CP3Calculator):
    """PAPER_228: Westerlund 2 â€” 9-term MUGE, rho_wind=1e-20 (10x LMC), OB-supergiant winds"""
    # Session 58 â€” grok_share_8d951e12.txt Doc 6
```

Located in CondensedPhysics3.py (Session 58).

6. Conclusion

Westerlund 2 establishes the upper bound of the MUGE stellar wind family with $\rho_{\text{wind}} = 10^{-20}$ kg/m³, placing it 10 $\text{\AA}$ — above the LMC Tapestry. The shared a_{wind} formula with different physical parameters demonstrates the parametric flexibility of the MUGE wind extension while remaining anchored to observationally motivated values.

Source: grokshare8d951e12.txt â€” Doc 6 (Westerlund 2 OB Wind MUGE)

UQFF computed: MUGE buoyancy ratio $U_{bi}/F_U = [SSq] \times r^2 / GM = 5.7e-1 \times 5.0e-4 = 2.85e-4$; compressed MUGE baseline $g = 5.4e-7 \text{ m/s}^2$ at r_{ISCO} .